

Bern 16

Sixteen mono channels into a stereo mix bus, with a broadcast console's clean summing amp and output transformer.



Overview

Bern 16 is a sixteen-channel summing mixer in the spirit of the classic Swiss broadcast consoles: precise and quiet, with a tight discrete-transistor summing amp and an output transformer that adds warmth when you lean on it. Each mono channel has its own gain, pan, solo, mute, meter and name strip, and the sixteen channels are mixed down to a stereo pair.

The colour comes from the master section. DRIVE pushes the mix into the summing amp and the transformer without making it louder: the amp rounds the peaks gently with a little 2nd and 3rd harmonic, and the transformer thickens the lows below about 100 Hz. COLOUR adds the console's own response, TONE tilts it darker or brighter, and IRON adds a whisper of input-transformer saturation to every channel.

It is very clean at DRIVE 0: a 0 dBFS peak loses only about 0.2 dB in the amp. Turn up DRIVE when you want the console to be heard.

Reach for it for

- Summing drums, stems or a whole mix "through a desk" inside REAPER
- Gentle glue and weight on a drum or music bus
- A quick balance and pan of up to sixteen mono sources in one place

- Darkening or brightening a mix by a hair with the console tilt
- Checking the balance in mono

Loading and setting it up

1. Put Bern 16 on a bus track and set that track to 16 channels.
2. Send each mono source to the bus, and in each send's audio routing pick a single channel, 1 to 16, so every source lands on its own channel strip.
3. Turn off each source's own master / parent send, so you only hear it through Bern 16.
4. The stereo mix leaves Bern 16 on channels 1 and 2; channels 3 to 16 are silenced on the way out.
5. Click a channel's scribble strip and type a name so you know what is where.

Controls

The header

PRESETS	Opens the preset list. The last item, Reset all to defaults, puts every channel and the master section back to the start.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. A new instance opens in L.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Channels 1 to 16

Sixteen identical mono strips, in groups of four. Channel 1 is the input on track channel 1, and so on.

Scribble strip	The white strip under the channel number. Click it and type a name (up to 16 characters); Enter keeps it, Escape cancels, Backspace deletes. Clicking elsewhere also keeps it. The names are saved with the project.
GAIN	The channel level, -24 to +24 dB (0 dB by default).
PAN	Left to right, L100 to R100 (C by default). Constant power: a centred channel is 3 dB down on each side, so it keeps the same loudness as you pan it.
Meter	A 12-segment level meter, 0 to -48 dBFS, read after GAIN and IRON. Green, amber from -9 dBFS and red from -3 dBFS. It goes dark when the channel is muted or not heard because of a solo.
S (SOLO)	Solo in place. Any number of channels can be soloed at once; then only the soloed channels are heard.
M (MUTE)	Mutes the channel. A muted channel stays muted even when it is soloed.

MASTER

DRIVE	Level into the summing amp and the output transformer, 0 to +18 dB (0 by default). It is level-matched, so it changes the colour, not the volume: more DRIVE gives rounder peaks, a little more harmonic richness and thicker lows.
TONE	The console's tilt, DARK 100 to BRIGHT 100 (FLAT by default). At full DARK the lows (shelf at 250 Hz) come up 2 dB and the highs (shelf at 4 kHz) go down 2 dB; BRIGHT does the opposite. It works only while COLOUR is in.
COLOUR	Switches the console response in or out (in by default): a gentle high-pass at the very bottom, a little transformer weight around 45 Hz, a soft top roll-off (about 1 dB down at 20 kHz) and the TONE tilt. Off is flat; the amp and transformer still work.

IRON	A whisper of input-transformer saturation on every channel (off by default). Mostly felt on hot channels: it softens them a little and adds a touch of 2nd harmonic.
MONO	Folds the mix to mono, to check the balance (off by default).
SOLO light	Flashes while any channel is soloed.
VOLUME	Master output level, -24 to +12 dB (0 dB by default).
DRIFT	How much the power supply wanders, 0 to 100% (20% by default). See Drift below.
SAT	Band saturation on the way out, 0 - 100% (0 by default): only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
PEAK / RMS meters	Left and right output meters, 0 to -54 dBFS. The segments show the peak in 2 dB steps, with a hold that falls back after 1.5 seconds; the thin blue bar beside each shows the RMS level (300 ms). The figures below show the held peak. Click the meters to clear the hold.

Drift

DRIFT gives the console an uneven power supply. Each channel's gain wanders on its own by up to 0.25 dB (at 100%), and the bias of the summing amp wanders separately on the left and right, so the colour shifts very slightly between the sides. The movements are slow and smooth. At 0% everything stays exactly where you set it.

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click to reset a knob to its default.

Presets

Presets 1 to 8 set the master section (DRIVE, TONE, COLOUR, IRON, VOLUME, MONO and DRIFT) and leave the channels, SAT and the output stage alone. Stereo Pairs and Wide Fan only set the pans; Mono Check only switches MONO on.

Init	The default master: no drive, flat tone, COLOUR in, IRON out, DRIFT 20%.
Clean Bus	A straight sum: COLOUR out and no drift.
Warm Iron	+6 dB of drive, a little darker, IRON in.
Drum Bus Glue	+9 dB of drive, a touch brighter, IRON in, VOLUME -1 dB.
Pushed Console	+14 dB of drive with IRON in, VOLUME -2 dB and a bit more drift: the console leant on hard.
Dark Broadcast	Light drive with the tone well over to DARK.
Bright Lift	A little drive with the tone towards BRIGHT.
Tired Supply	+7 dB of drive, slightly dark, IRON in, DRIFT at 85%: an old desk on a wandering supply.
Stereo Pairs	Pans the channels as stereo pairs: odd channels hard left, even channels hard right (1/2, 3/4 ... 15/16).
Wide Fan	Spreads the channels evenly from channel 1 hard left to channel 16 hard right.
Mono Check	Switches MONO on and changes nothing else.
Reset all to defaults	Every channel back to 0 dB, centre, unmuted and unsoloed, the master to Init, and SAT to 0. The channel names are kept.

Tips

- Use Stereo Pairs when you feed stereo sources in as two mono channels (left into an odd channel, right into the next even one).
- DRIVE is level-matched, so judge it by the shape of the sound, not the loudness. Watch the peak meter: more drive brings the peaks down a little.
- For the cleanest sum, load Clean Bus; for a real "through the desk" feel, try Warm Iron and adjust DRIVE to taste.

- Name the channels as you route them; the names stay with the project.
- Set DRIFT to 0 for a static, repeatable mix; leave a little in for a less clinical feel.

File: gfx-bern-16.jsfx